

Final Revision Reference Card

92%

Estimated readiness | Pass threshold: 72% | Buffer: +20 points

Readiness by domain

Domain	Weight	Readiness	Watch area
Agentic architecture	27%	82%	Critical path timing, handoff vs checkpoint
Claude Code + MCP	38%	88%	--print flag, \$1 syntax, server permission scope
Prompt engineering	20%	96%	Retry loop correction prompt, 4 system prompt components
Context management	15%	95%	Over-aggressive compression, goal drift vs constraint loss
Overall	100%	92%	Above 72% pass threshold in every domain

The 8 most common exam traps

These traps appeared repeatedly across all 18 days. Memorise every one.

- 1 --ci and --headless flags don't exist**
The only CI flag is --print. The exam offers --ci, --headless, --auto as distractors. It is always --print, every time.
- 2 \$1 doesn't exist in slash command files**
Slash commands are markdown natural language. No \$1, {arg}, or %s. Arguments are described in prose: 'Review the file provided as the argument.'
- 3 'Goal explicitly authorised it' is never sufficient**
Goals are written before the agent knows what it will find. Category authorisation ≠ instance authorisation. Irreversible actions always need explicit human confirmation.

■ 4 **Parallel time = MAX, not sum or average**

Sequential total = SUM. Parallel total = MAX(parallel agents) + SUM(sequential steps after). The exam hides a sequential step after the parallel phase — always check.

■ 5 **Trusted subagent ≠ trusted data**

Trust applies to the subagent's execution, not to the data content it returns. All subagent outputs, MCP tool results, and web content are untrusted input.

■ 6 **Required → Optional = SAFE (not breaking)**

Making a required field optional is MORE PERMISSIVE = safe change. Optional → Required is breaking. Adding optional fields is safe. Removing any field is breaking.

■ 7 **Multiple problems = check for combined answer**

When you spot one problem in a scenario, ask 'what else is wrong?' before selecting. Combined answer options (e.g. 'Both B and C') are frequently the correct choice.

■ 8 **Self-review doesn't fix shared blind spots**

Both passes of self-review share the same reasoning errors. For high-stakes decisions (medical, legal, financial), multi-instance review is required.

Key formulas — memorise these cold

Critical path timing

```
Total = MAX(parallel agents) + SUM(sequential steps after)
```

```
Example: 3 agents (4s, 9s, 6s) + synthesis (3s)
```

```
= max(4,9,6) + 3 = 9 + 3 = 12 seconds
```

```
Trap: parallel max LOOKS like the answer – check for sequential steps after
```

Context compression trigger

```
Compress at ~70% context usage – NOT at 90% or 100%
```

```
At 95%: truncation has ALREADY happened – original goal may be gone
```

```
At 70%: you still have time to preserve everything critical
```

Validation retry loop

```
MAX_RETRIES = 3 (never more – 10 is the anti-pattern)
```

```
Retry 0: original prompt
```

```
Retry 1+: correction prompt WITH failed output included
```

```
After max: LOG + ESCALATE + RAISE ERROR (never continue silently)
```

CLAUDE.md precedence

Loading order: Global → Project → Local

Precedence: Local > Project > Global (most specific wins)

NOTE: Loading order and precedence are OPPOSITE — a common trap

Multi-instance aggregation

All agree → high confidence → auto-return result

Partial disagree → medium confidence → flag for human review

Any disagree → (medical/legal/financial) → escalate ALWAYS

Majority vote → NEVER appropriate in high-stakes domains

MCP schema change rules

More permissive (loosening) = SAFE change

More restrictive (tightening) = BREAKING change

Required → Optional = loosening = SAFE ✓

Optional → Required = tightening = BREAKING ✗

Add optional field = loosening = SAFE ✓

Remove any field = tightening = BREAKING ✗

10 concepts that separate passing from failing

1 Three MCP tool boundary violations

Over-broad (one tool, too many capabilities) → split by operation. Overlapping (two tools doing same thing) → merge or differentiate. Over-permissioned (excess credentials) → minimal footprint.

2 Four CLAUDE.md levels and rules

Global=personal/never commit/lowest. Project=team/always commit/middle. Local=subdirectory/highest. All three load and merge — conflicts resolve by specificity (local wins).

3 Four required system prompt components

Role definition + Constraints/rules + Output format specification + Escalation conditions. Missing any one produces unpredictable production behaviour.

4 CI/CD four-point checklist

--print flag + JSON output ('ONLY valid JSON, no preamble') + Exit code instruction + API key from secrets. All four must pass — if any fails, the option is wrong.

5 Lost-in-the-middle fix toolkit
Chunk documents into sections + Strategic positioning (critical content at start/end) + Instruction repetition (repeat key constraints at both start and end).

6 Self-review vs multi-instance
Self-review: moderate stakes, cost-sensitive, ~15-25% improvement, shares blind spots. Multi-instance: high stakes, independent instances, no shared blind spots. Medical/legal/financial: always multi-instance.

7 Three MCP error types
Schema error → self-correct input, retry once. Transient (rate limit, timeout) → exponential backoff, cap at 2-3 retries. Permanent (auth failed, 404) → fallback or escalate, NEVER retry.

8 Handoff vs checkpoint vs compression
Compression: context too full → compress in-place → same agent continues. Handoff: pass state to fresh agent → clean context window. Checkpoint: save to external storage → crash recovery. Three different tools, three different problems.

9 Minimal footprint — applied everywhere
Tools: minimum permissions. Agents: minimum capabilities. Actions: human confirmation before irreversible. CI prompts: explicit authorisation in the prompt. 'The goal authorised it' is never sufficient for irreversible actions.

10 Sequential vs parallel vs hybrid
Data/state/order dependency → sequential. No dependency → parallel. Parallel within stages + sequential between stages → hybrid. Trigger words: 'ETL/pipeline/stages'=sequential, 'independent/batch/fan out'=parallel, 'within each stage'=hybrid.

Exam-day strategy

60 questions in 120 minutes = 2 minutes per question. Your target score is 720/1000 — roughly 44 correct out of 60. You've been scoring 95%+ on 20-question mocks. You have significant buffer.

First pass — answer all easy questions

- Don't spend 5 minutes on a hard question when 3 easy ones are waiting. Flag hard questions and move on. Complete all easy questions first.

Read ALL options before selecting

- The exam puts a 'mostly right' option before the 'completely right' option. Read all four before committing. Never stop at the first plausible answer.

Watch for the combined answer trap

- When you spot one problem, ask 'what else is wrong?' before selecting. Combined options (Both B and C) are frequently the correct choice in two-problem scenarios.

Eliminate distractors first

- The exam always includes 1-2 obviously wrong options. Eliminating them makes the choice between the remaining 2 much easier and faster.

Trust domain keywords

- 'parallel'=MAX timing. 'crash'=checkpoint. 'middle of document'=chunking. 'medical/legal/financial'=escalate on any disagreement. Keywords are your shortcuts.

Never leave blank

- No penalty for wrong answers. If stuck, eliminate 2 options and guess between remaining 2. You're right 50% of the time. Blank is always 0%.

Complete course summary

Week	Days	Domain	Score
Week 1	1-5	Agentic architecture (27%)	78%
Week 2	6-10	Claude Code + MCP (38%)	72%
Week 3	11-13	Prompt eng + Context (35%)	100%
Week 4	14,17	Mock exams	97.5% (39/40)
Overall	18 days	All domains	92% readiness

You are ready. Go pass the CCA-F. ■